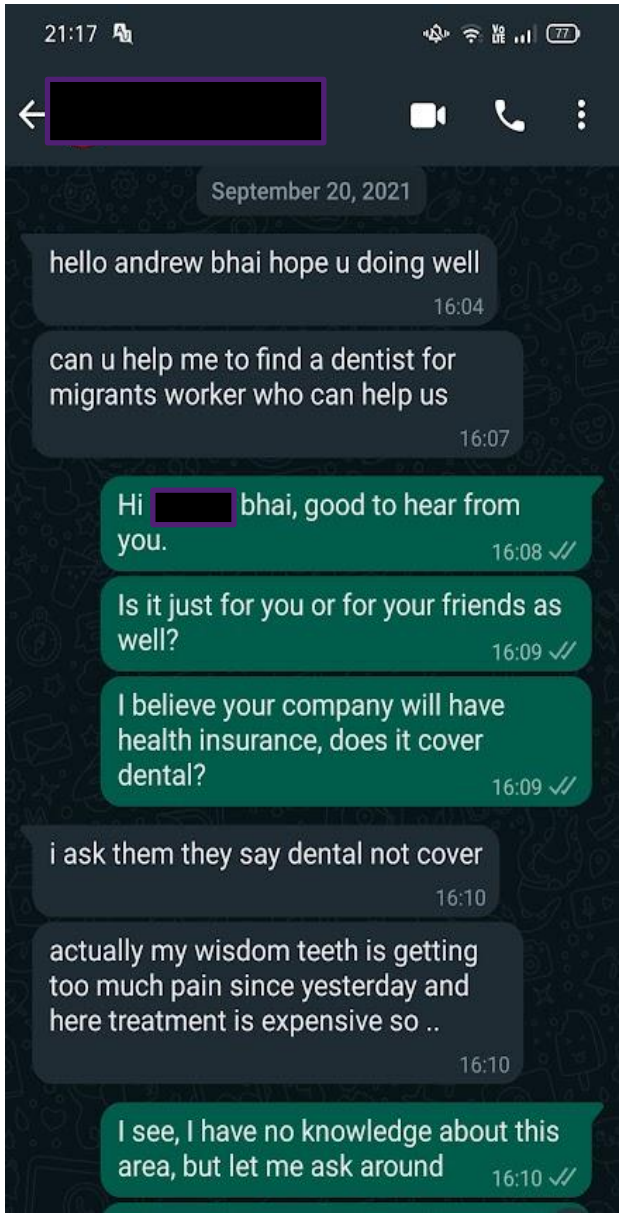
PROJECT DETAILS

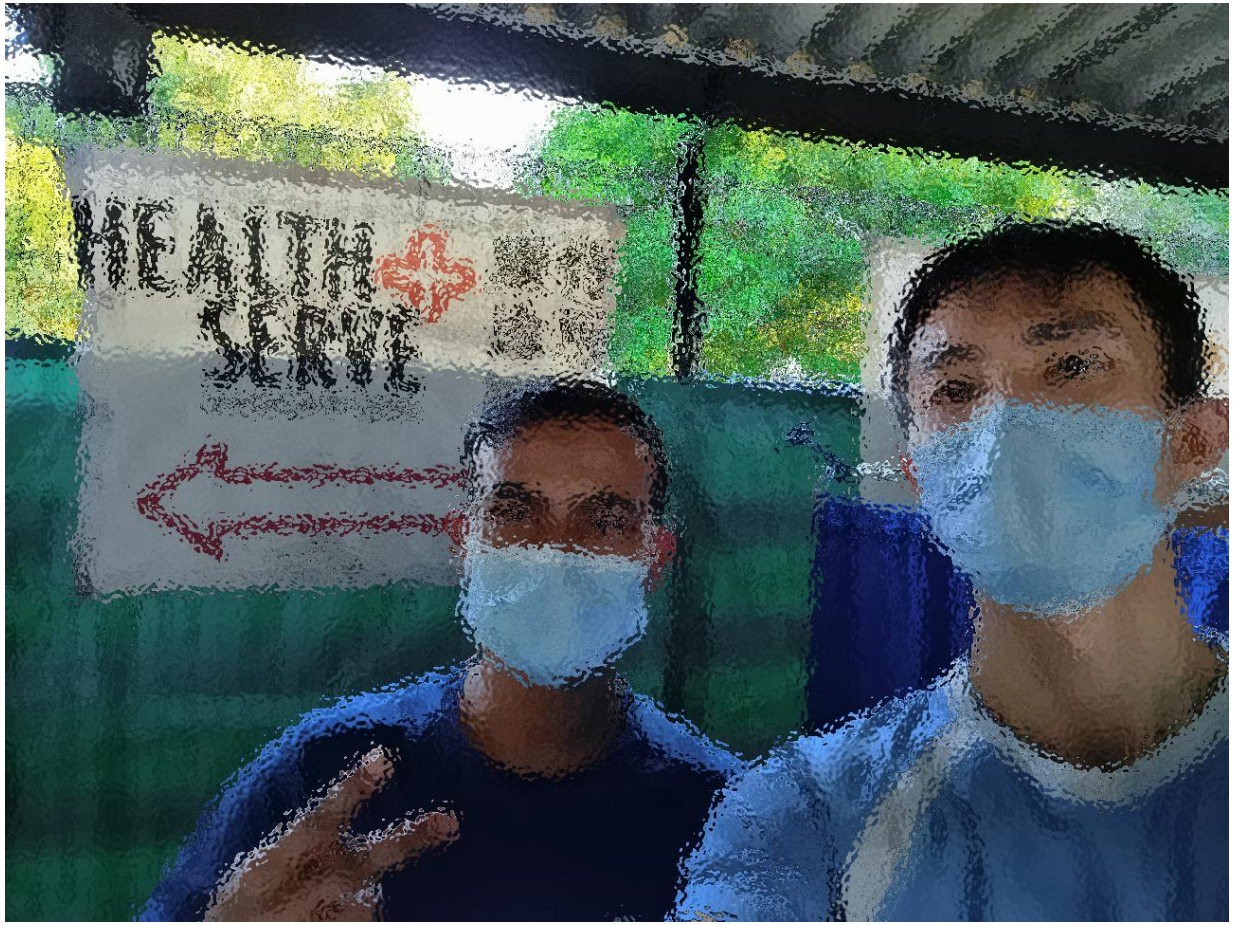
What comes to mind ?



Picture: https://allegiance.com.sg/

The genesis


21:17
[redacted]
September 20, 2021
hello andrew bhai hope u doing well
16:04
can u help me to find a dentist for migrants worker who can help us
16:07
Hi [redacted] bhai, good to hear from you.
16:08
Is it just for you or for your friends as well?
16:09
I believe your company will have health insurance, does it cover dental?
16:09
i ask them they say dental not cover
16:10
actually my wisdom teeth is getting too much pain since yesterday and here treatment is expensive so ..
16:10
I see, I have no knowledge about this area, but let me ask around
16:10



AI raises the premium on empathy-driven engineering

Times Higher Education | Inside Higher Ed

‘AI raises the premium on empathy-driven engineering’

As knowledge becomes increasingly accessible, engineering students will need deep understanding and intrapersonal intelligence to make product-design decisions with end users in mind

STEM teaching and research | Artificial intelligence in higher education | Higher education pedagogy | Feature article | Asia



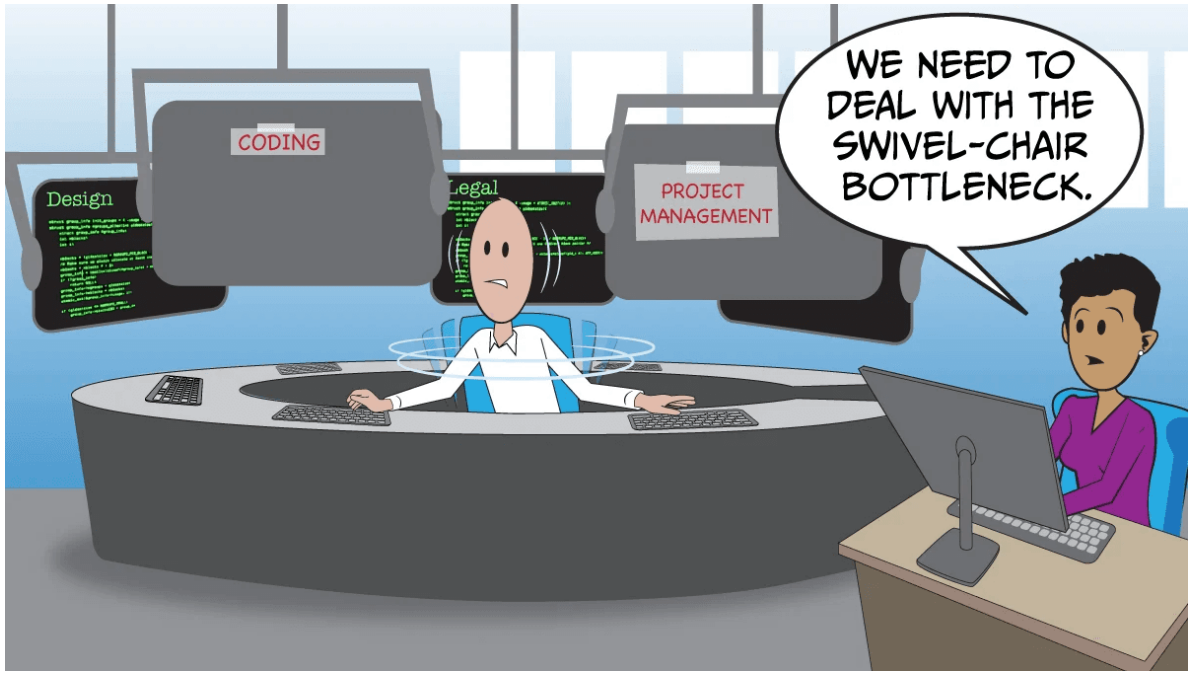
Andrew Koh
Singapore Management University

14 Jul 2026

AI and low-/no-code tools compress routine coding, but deep understanding remains essential and empathy becomes engineering

Ref: https://www.timeshighereducation.com/campus/ai-raises-premium-empathydriven-engineering

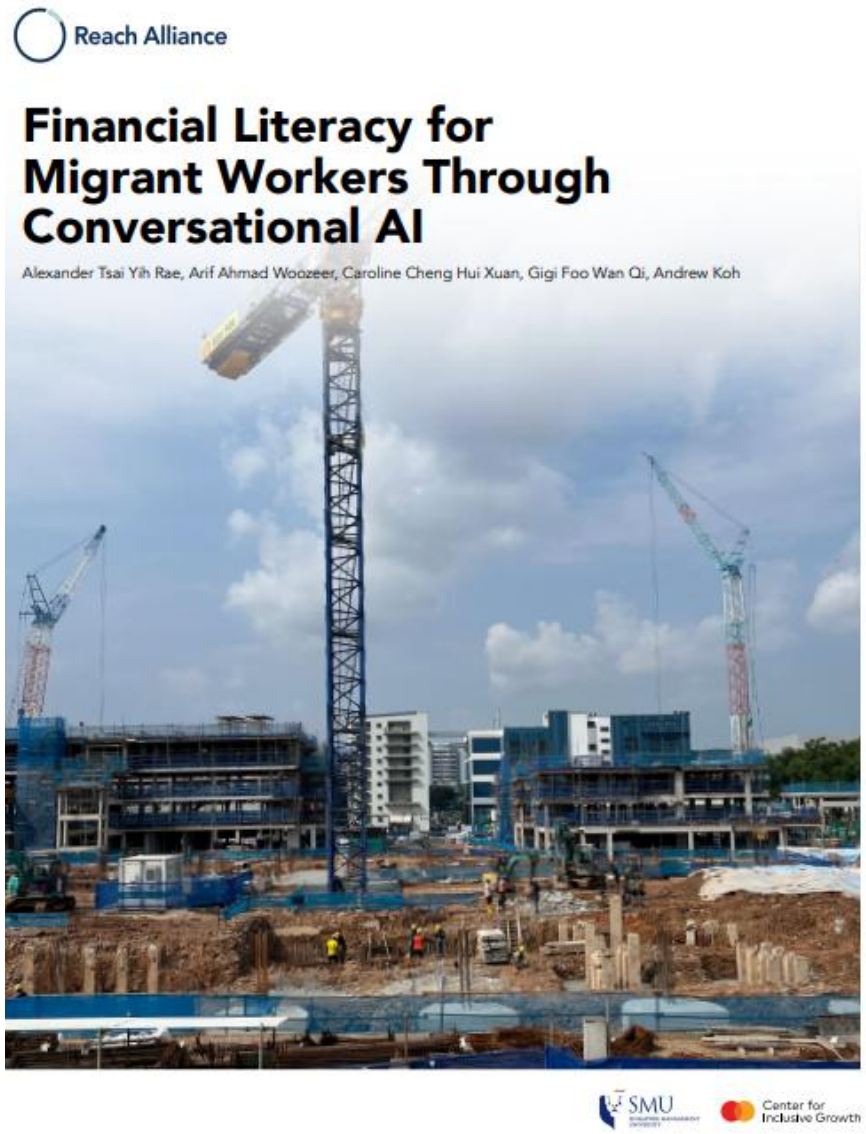
Rise of product management



1. Accelerated development times shifts bottleneck to product management.
2. “The tech industry has more engineers than PMs, but both are promising paths. If you are an engineer, you’ll find it useful to learn some product management skills, and if you’re a PM, please learn to build!”

   – Andrew Ng, Founder of DeepLearning.AI

Ref: https://www.deeplearning.ai/the-batch/ai-native-software-development-needs-generalists



Key lessons

1. Context must come before advice
2. Build with people, not merely for them

Ref: https://reachalliance.org/reports/how-can-we-utilise-conversational-a-i-to-enhance-financial-literacy-for-migrant-workers-in-singapore/

Lesson 1: Context must come before advice

One size fits no one: One worker earned $600-$800monthly and had nothing left after debt, remittances,food and transport. A fixed $50 target was not motivating-it was unrealistic..

Money is often collective: Financial choices included awife's needs, a sibling's education, marriage and buildinga family home. Advice centred only on the individualmissed the real decision-making unit..

Knowledge already existed: Workers were not emptyvessels lacking financial wisdom. Many had aspirationsand informal strategies; they needed tools that couldorganise and support practices they already used.

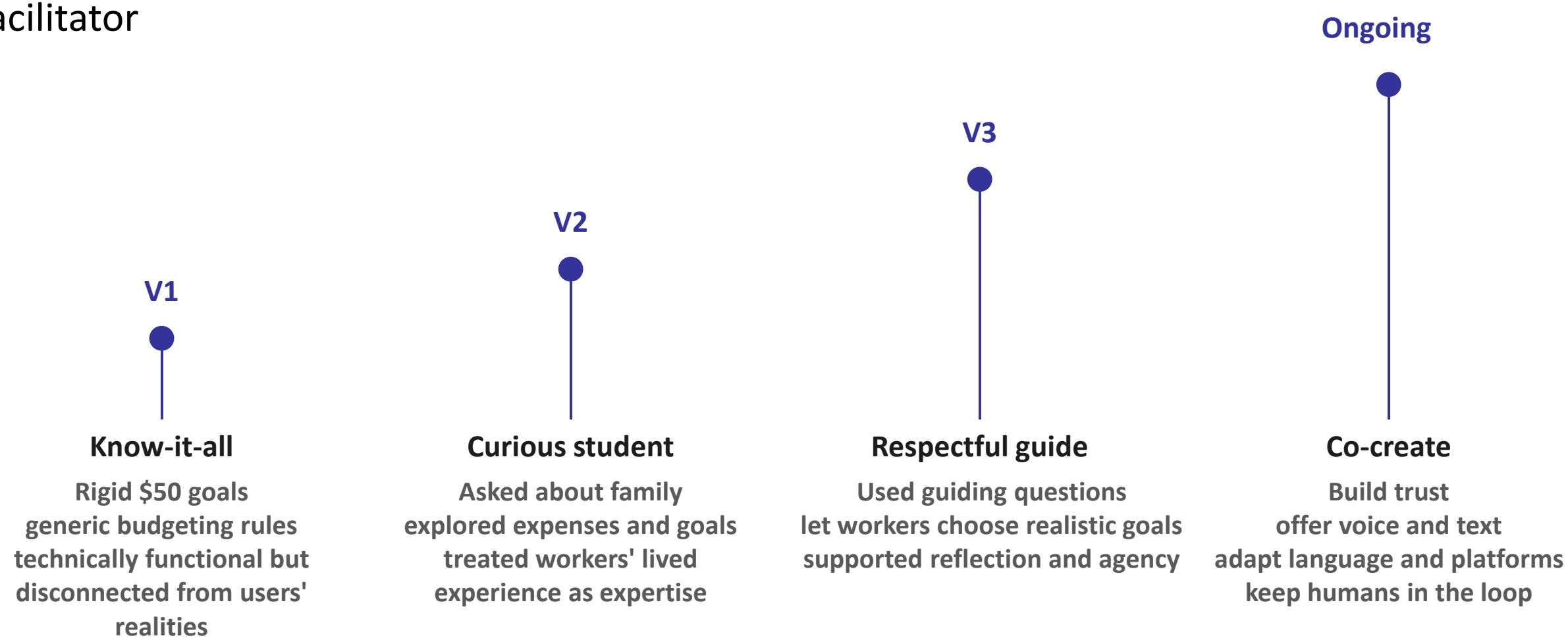
Ask before recommending: Students redesignedonboarding to learn about income, expenses,responsibilities and goals first-so guidance could fiteach worker's actual constraints and priorities.



Ref: https://reachalliance.org/reports/how-can-we-utilise-conversational-a-i-to-enhance-financial-literacy-for-migrant-workers-in-singapore/

Lesson 2: Build with people, not merely for them

Student learning reshaped the chatbot from prescriptive expert to contextual, respectful facilitator



Ref: https://reachalliance.org/reports/how-can-we-utilise-conversational-a-i-to-enhance-financial-literacy-for-migrant-workers-in-singapore/

Project Overview

Using conversational AI to address societal challenges

Societal challenges such as climate change, mental health, inequitable access to information, lack of educational resources, etc, continue to impact communities worldwide. These challenges are often exacerbated by limited human resources, language barriers, and a lack of scalable solutions.

Conversational AI presents a unique opportunity to address these issues effectively. By leveraging advancements in natural language processing (NLP) and machine learning, conversational AI can provide accessible, scalable, and personalized support to communities, enabling individuals to navigate challenges and access opportunities more effectively.

Project Overview: Potential Use Cases

Migrant Workers: Provide financial literacy for migrant workers

- Simplify complex financial concepts through interactive conversations.
- Assist with budget planning, expense tracking, and saving goals.
- Provide access to microfinance options or guide users through application processes for loans or grants.

Sustainability and Environmental Advocacy

- Help organizations with sustainability reporting
- Enhance one's nature connectedness
- Educate users on eco-friendly practices, such as reducing waste or conserving energy.
- Provide tips on sustainable living and connect users with local environmental initiatives.

Mental Health

- Serve as a first point of contact for mental health concerns, providing empathetic conversations and suggesting coping strategies.
- Direct users to professional help through resources and helplines.
- Offer mood tracking and personalized recommendations for improving well-being.

Education and Skill Development:

- Deliver interactive learning modules, quizzes, and practice exercises for various subjects or skill areas.
- Act as a virtual tutor, answering questions and providing explanations in real-time.
- Offer personalized career counseling and guidance on academic or professional pathways.

Assessment

Assessment of your chatbot will be based on the following five areas:

a. Audience analysis and research (30%)

i. Articulation and elaboration of use cases and problem statements
ii. Identification of pain-points and gains (Use empathy map)
iii. Relevant examples or potential methods relating to problem statement
iv. Research sources can include academic literature, articles, news, etc. (Tip: For academic literature, consider AI tools such as Ai2 Paper Finder, Scispace and Undermind)

b. Design and development of solution (20%)

i. How does the solution solve the problem statements
ii. Design of conversation flows
iii. Experimentation of methods to improve accuracy of responses

c. Testing (20%)

i. Record, show and explain the process of both internal testing and field testing by users outside the team
ii. Scope should include 1) flow of conversation 2) accuracy and relevance of responses and 3) others
iii. If multiple tools or methods are used, how does the performance vary across
iv. Documentation of iterations/improvements made to solution

d. Transparent and critical use of GenAI (10%)

e. Peer Evaluation (20%)

Grading Rubric (Audience analysis and research)

Criterion	Excellent	Good	Satisfactory	Needs Improvement
Problem definition	Defines a specific, meaningful, and well-scoped problem for a clearly identified user group. Explains why the problem matters using evidence.	Defines a clear problem and user group, with some supporting evidence.	Problem is generally stated but may be broad, vague, or weakly linked to users.	Problem is unclear, overly broad, or not connected to a real user need.
User understanding and empathy	Demonstrates deep understanding of users' needs, constraints, emotions, motivations, and context. Uses tools such as empathy maps, personas, journey maps, or interviews effectively.	Shows good understanding of users with some consideration of context and pain points.	Identifies some user needs or pain points but lacks depth or supporting evidence.	Little or no evidence of user understanding. Pain points are assumed rather than researched.
Research quality	Uses a strong range of credible sources, including academic literature, reports, articles, case studies, or interviews. Sources are clearly connected to project decisions.	Uses credible sources and connects most research to the project.	Uses limited sources or sources are only loosely connected to the project.	Uses few, weak, irrelevant, or uncited sources.
Insight extraction	Clearly translates research findings into design implications for the chatbot.	Provides some useful insights that inform the chatbot design.	Research findings are summarized but not clearly translated into design decisions.	Research is descriptive only, with little or no impact on design.
Use case clarity	Presents realistic chatbot use cases, including user scenarios, goals, and expected value.	Presents clear use cases with minor gaps in realism or detail.	Use cases are basic or generic.	Use cases are missing, unclear, or not feasible.

Grading Rubric (Design and Development of Solution)

Criterion	Excellent	Good	Satisfactory	Needs Improvement
Alignment with problem statement	Solution directly addresses the identified problem and user needs. Design choices are clearly justified using research findings.	Solution addresses the problem well, though some design choices need stronger justification.	Solution addresses part of the problem but has gaps or weak alignment with user needs.	Solution does not clearly address the stated problem.
Conversation flow design	Conversation flows are logical, user-centric, context-aware, and include onboarding, main task flow, fallback handling, and closure.	Conversation flows are mostly logical and functional, with some minor gaps.	Conversation flows are basic and may lack context-gathering, fallback responses, or smooth transitions.	Conversation flows are confusing, incomplete, or not user-centered.
Context-awareness and personalization	Chatbot gathers relevant context before giving advice and adapts responses to user needs, constraints, or preferences.	Chatbot gathers some user context and provides partially tailored responses.	Chatbot gives mostly generic responses with limited personalization.	Chatbot provides generic or inappropriate advice without understanding the user's situation.
Prototype quality	Prototype clearly demonstrates key user journeys and is easy to understand, use, and evaluate.	Prototype demonstrates main functions but may lack polish or completeness.	Prototype demonstrates limited functionality or only partial user journeys.	Prototype is missing, non-functional, or difficult to evaluate.
Experimentation and improvement methods	Team systematically experiments with prompts, tools, knowledge bases, response structures, or model settings. Comparisons are documented with clear findings.	Team experiments with some methods and provides basic comparisons.	Limited experimentation; changes are made but not systematically documented.	Little or no experimentation is shown.

Grading Rubric (Testing)

Criterion	Excellent	Good	Satisfactory	Needs Improvement
Testing plan	Provides a clear testing plan covering objectives, participants, test scenarios, methods, and success metrics.	Provides a testing plan with minor gaps in scope or metrics.	Testing plan is basic and lacks clear metrics or participant details.	Testing is ad hoc, unclear, or missing.
Internal testing	Conducts structured internal testing across conversation flow, response accuracy, relevance, tone, safety, and fallback handling.	Conducts internal testing across most important areas.	Conducts limited internal testing with few scenarios.	Little or no internal testing is documented.
Field/user testing	Tests the chatbot with users outside the team and captures meaningful feedback. User feedback clearly informs improvements.	Tests with some external users and captures useful feedback.	Limited external testing or feedback is shallow.	No external user testing or feedback is provided.
Response quality evaluation	Evaluates accuracy, relevance, clarity, empathy, safety, and usefulness of chatbot responses using clear criteria.	Evaluates most major aspects of response quality.	Evaluates response quality informally or only partially.	No meaningful evaluation of response quality.
Tool or method comparison	If multiple tools, prompts, or methods are used, compares their performance and explains trade-offs.	Provides some comparison with basic insights.	Mentions different tools or methods but provides little analysis.	No comparison or reflection is provided.
Iteration documentation	Clearly documents what changed between versions, why changes were made, and how testing informed improvements.	Documents main iterations with some explanation.	Mentions improvements but lacks detail on evidence or rationale.	Iterations are missing or poorly documented.

Grading Rubric (Transparent and critical use of GenAI)

Criterion	Excellent	Good	Satisfactory	Needs improvement
Process transparency	Clearly identifies the GenAI tools/models used, their purpose at each project stage, relevant prompts or workflows, and how outputs were modified or incorporated. Appropriate attribution is provided and sensitive data is protected.	Identifies the main tools, purposes and workflows. Most uses and modifications are documented, with minor gaps.	Lists the tools used but provides limited explanation of their purpose, workflow or influence on the final work.	GenAI use is poorly documented, unclear or not disclosed. There is little evidence of appropriate attribution or data protection.
Critical evaluation and verification	Systematically verifies GenAI outputs using credible sources, testing or user evidence. Clearly identifies limitations, errors and biases, and documents rejected or revised outputs. Demonstrates strong student judgment.	Verifies most important outputs and identifies relevant limitations. Decisions to accept or revise outputs are generally justified, with minor gaps.	Some verification is attempted, but it is inconsistent or weakly documented. Reflection on limitations and biases is basic.	GenAI outputs are accepted largely at face value. Verification, critical reflection and evidence of student judgment are minimal or absent.

Deliverables

Deliverable 1: Audience analysis and research handout (10% of total project score, submit by end of Week 7)

Deliverable 2: Group presentations in week 13

a. Your presentations should be structured around the areas of assessment:
   a. Research and Problem Analysis
      - Who is the audience and what is their problem?
      - What does the research tell us?
      - So, what should AI do?
   b. Design and Development of Solution: Explain how your solution addresses the problem, showcase conversation flows, and present prototypes.
   c. Testing: Describe testing processes, analyze results, and outline improvements made.
   d. Transparent and critical use of GenAI

a. Include a functional link to your conversational agent in the presentation, ensuring it is accessible and demonstrates the progress of your work

b. Each presentation will be for 10 minutes and should not contain more than 15 slides

Deliverables

Deliverable 3: Working files

Before your presentation in week 13, each group will be granted access to a folder where you will upload all your working files. This provides a chance for you to showcase the work that your group has done.

Please upload your slides and working files by 3pm the day before your presentation.

Examples of content to include:

a. Research Files: Audience analysis and research handout, other relevant research files
b. Design Documents: Conversation flow diagrams, and other design-related materials.
c. Testing Documentation: Records of internal and field testing, including test cases, results, and user feedback.
d. Iteration Logs: Detailed records of improvements or changes made between iterations.
e. Any other relevant content.

(Past examples) MOM's Settling-in Programme (SIP)

The SIP is a 1-day orientation programme to educate migrant workers on Singapore's social norms, their employment rights and responsibilities, Singapore laws and where and how to seek assistance.

Aim: Developing a chatbot in the native language (start with English) of MWs to offer round-the-clock assistance and access to reliable information, along with the following benefits:

1. Immediate Assistance: MWs can promptly access information without waiting for human support.
2. 24/7 Availability: Available at all times, providing support beyond regular business hours (related to the above).
3. Scalability: Capable of handling numerous inquiries simultaneously.
4. Empowerment: MWs can independently find answers to their queries, enabling them to resolve issues autonomously.
5. Cost-Effectiveness: Potential reduction in the need for human customer support agents, freeing up resources for more complex tasks. (related to the above 3)

(Resources) This table contains the topics found in SIP and provides the reference to where the content is found.

(Past examples) Financial Literacy for migrant workers

Digital literacy is crucial for migrant workers to access internet banking services. Your project is based on years of ongoing work by DBS to provide related training for migrant workers to bridge gaps related to digital literacy, language barrier and security concerns.

Aim: Develop a chatbot to enhance digital financial literacy for migrant workers

Focus areas:

1. Usage of Digibank app
2. Heightened awareness of scams and phishing messages

Resources

1. POSB Website: Payroll Account For Work Permit Holders
2. DBS Help and Support Page
3. Digibank user guides (English, Bahasa Indonesia and Tagalog)
4. Relevant parts of MW Handy Guide

(Past examples) Nature connectedness

Changing consumer behavior plays a central role in sustainability. Connectedness to nature can significantly increase awareness and action towards more eco-friendly practices.

Problem statement:

How can we design chatbots that create or facilitate "fitting" nature connectedness interventions?

Considerations:

1. Your target audience
2. What context/type of experience the activity is designed for: direct/vicarious?
   - Experiences based in Singapore/ Featuring Singapore's nature are encouraged.
3. What information to collect so that the chatbot can provide recommendations that "fit" the users:
   - Personality? Strengths? Preferences?
4. What aspects of the activity the AI chatbot can customize (e.g., multimedia elements).
   - Will you handpick a library of content for the chatbot to present, or allow the chatbot to create/source its own content?